

How long do you need?

Ground-Based Follow-up of LSST Transients — Hands-on Assignment

CASSA TDMMA Workshop 2026
Independent University, Bangladesh
Instrument: CRITO driving a simulated observatory
300 mm, 5.2 μm , INDI simulators

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1 Setup

1.1 What you are flying

You will drive a complete observatory control system — CRITO — against *simulated* hardware. Every step is the real code path: the same INDI drivers, the same plate solver, the same exposure planner, the same archive writer. Only the photons are fictional.

Role	Driver you connect to	What it presents
Mount	INDI Telescope Simulator	300 mm optics, slews and tracks
Imaging camera	INDI CCD Simulator	5.2 μm pixels, star field from gsc
Guide camera	INDI CCD Simulator (guide role)	a second chip for PHD2
Focuser + wheel	INDI simulators	motorised focus, L/R/G/B/Ha/OIII/SII
Roof	INDI Dome Simulator	opens and closes
Broker feed	ALeRCE (ZTF, LSST-style)	live poll, or cached

Optical specification

Focal length	300 mm
Pixel size	5.2 μm
Filters	L, R, G, B, Ha, OIII, SII
Sub-exposure cap	300 s

These are the values used both on paper and in the Exposure tab.

1.2 Participant setup (five minutes, at your seat)

1. Open an SSH tunnel to the workshop server. The instructor will give you an SSH USER, SSH HOST, and password during the session. Open Terminal (macOS/Linux) or PowerShell (Windows) and run:

```
ssh -N -L 8000:127.0.0.1:8000 SSH_USER@SSH_HOST
```

Replace SSH_USER and SSH_HOST with the values provided. The first connection may ask whether you trust the host key; confirm it only after the instructor verifies the fingerprint. Enter the session password when prompted. Password typing is intentionally invisible. Leave this terminal open for the entire exercise. You can stop the tunnel afterwards with **Ctrl+C**.

What the tunnel does. It maps port 8000 on your own computer to port 8000 on the workshop server. After it connects, open <http://localhost:8000> in your browser. The application is still running on the server; localhost now reaches it through the encrypted SSH connection. If local port 8000 is already occupied, use `-L 8080:127.0.0.1:8000` and open <http://localhost:8080> instead.

2. Claim your telescope.

1. Open <http://localhost:8000> through the tunnel above.
2. Log in with your workshop account: `workshopN` / `tdmma-2026`.
3. **Confirm the simulation badge** in the top bar. If it is not there, stop: you are pointed at real hardware and your account will (correctly) refuse to move anything.
4. On **Locations**, find your assigned telescope (`workshop1` → **Simulator 1**, and so on) and click **Operate**. Other assigned telescopes are disabled; an occupied one says **Busy**. Your claim is exclusive while you work.
5. **Devices** → **Scan** → **Auto-detect & connect all**. Six roles should bind: mount, camera, guide camera, focuser, filter wheel, roof.

6. **Exposure** tab → confirm a calibration table is loaded (gains 0/60/120/200; filters L, R, G, B, Ha, OIII, SII). If it says *no calibration table*, tell your instructor.

What your account can and cannot do. The **workshop** role is ranked with **viewer** and promoted to operator-level *only on a backend that declares itself a simulator*. So the same login flies your simulator and is a spectator on the real telescope. Reads — telemetry, archive, atlas, candidates — are never blocked, because watching the real rig work is half the point. A refused command names its reason:

```
{"detail":"this account is limited to a simulation backend
  - this one drives the real observatory"} [403]
```

1.3 Pre-flight checklist

Check	If it fails
☐ SIMULATION badge in the top bar	wrong backend — do not proceed
☐ six devices connected after <i>Scan</i>	re-run auto-detect; check indiserver is up
☐ a test frame shows <i>stars</i> , not noise	gsc missing from the sim image
☐ Exposure tab lists gains and filters	CRITO_CALIBRATION_FILE unset
☐ Candidates tab has ≥ 1 target	lower CRITO_ALERCE_PROBABILITY , re-poll
☐ a plate solve succeeds	FOV hint vs. database size mismatch

2 The physics you need

Everything below is in **electrons**. This is the whole of the reference sheet; Part 1 needs nothing else.

2.1 Geometry

$$\text{plate scale: } s = 206.265 \times \frac{p [\mu\text{m}]}{f [\text{mm}]} \quad [\text{arcsec/px}] \quad (1)$$

$$\text{PSF width: } \theta_{\text{px}} = \frac{\text{seeing} [\text{arcsec}]}{s}, \quad \sigma = \frac{\theta_{\text{px}}}{2.3548} \quad (2)$$

$$\text{aperture: } n_{\text{pix}} = \pi (1.5 \theta_{\text{px}})^2 \quad (3)$$

A radius of $1.5 \times \text{FWHM}$ encloses about 99% of a Gaussian PSF's flux — that is the aperture the photometry is done in, and n_{pix} is how many pixels' worth of sky, dark and read noise you are forced to swallow along with the star.

2.2 Rates

$$\text{source: } S = 10^{-0.4(m-ZP)} \quad [\text{e}^- \text{s}^{-1}] \quad (4)$$

$$\text{PSF peak: } P = \frac{S}{2\pi\sigma^2} \quad [\text{e}^- \text{s}^{-1} \text{px}^{-1}] \quad (5)$$

The zero point ZP is defined as the magnitude that yields exactly $1 \text{ e}^- \text{s}^{-1}$ through the whole system — optics, filter, sensor. It is measured per filter, on-sky, at your site; it is *not* a property of the camera.

2.3 The CCD equation

For a single sub-exposure of length t :

$$\text{SNR}_{\text{sub}} = \frac{St}{\sqrt{\underbrace{St}_{\text{source shot}} + n_{\text{pix}}(\underbrace{Bt}_{\text{sky shot}} + \underbrace{Dt}_{\text{dark}} + \underbrace{R^2}_{\text{read}})}} \quad (6)$$

with B the sky rate ($\text{e}^- \text{s}^{-1} \text{px}^{-1}$), D the dark current ($\text{e}^- \text{s}^{-1} \text{px}^{-1}$) and R the read noise (e^- , *not squared* in the table — square it here).

2.4 The two bounds on a sub-exposure

$$\text{sky-limited floor: } t \geq \frac{k R^2}{B}, \quad k = 10 \quad (7)$$

$$\text{saturation ceiling: } t \leq \frac{0.7 \text{FW}}{P_{\text{brightest}} + B + D} \quad (8)$$

Equation (7) says: make the sub long enough that sky shot variance is at least ten times the read variance, otherwise you pay the read-noise toll on every single frame you stack. Equation (8) says: the brightest star you care about must stay in the linear part of the well — 70 % of full well is the usual safe fraction. A third bound is operational, not physical: **your mount and guider cap the sub** regardless of what the photons allow.

2.5 Stacking

$$\text{SNR}_{\text{total}} = \text{SNR}_{\text{sub}} \sqrt{N}, \quad N = \left\lceil \left(\frac{\text{SNR}_{\text{required}}}{\text{SNR}_{\text{sub}}} \right)^2 \right\rceil, \quad t_{\text{total}} = N t \quad (9)$$

$$\sigma_{\text{mag}} = \frac{1.0857}{\text{SNR}_{\text{total}}} \quad (10)$$

2.6 The calibration table

These are the constants the planner is loaded with (`calibration/minicam8.example.yaml`). Read noise and full well change with gain, because on a CMOS sensor they must; sky rate and zero point are per *filter* and per *site*, and are gain-independent.

This exercise uses the gain-60 row and the R row. The rest is here so you can check that the table on screen matches, and so you can see that sky rate and zero point are properties of the filter and the site, not of the camera.

Gain setting	Read noise R [e^-]	Full well FW [e^-]	System gain [e^-/ADU]
0	5.0	51 000	0.80
60	3.2	38 000	0.45
120	1.1	13 000	0.16
200	0.9	9 000	0.10

Filter	Sky B [$\text{e}^- \text{s}^{-1} \text{px}^{-1}$]	Zero point ZP
L	8.00	21.0
R	3.00	20.3
G	3.50	20.4
B	2.00	19.8
Ha	0.12	18.6
OIII	0.15	18.8
SII	0.10	18.3

Other

Pixel pitch p	5.2 μm
Dark current D at -10°C	$0.01 \text{ e}^- \text{ s}^{-1} \text{ px}^{-1}$
Focal length f	300 mm

3 The assignment

Part 1 — Plan it*25 minutes, on paper*

The situation. It is 19:40 on a clear September evening in Dhaka. An alert comes through the broker: a supernova candidate discovered two nights ago, currently at $R = 20.0$ and still rising. The follow-up coordinator sends the same message to every node in the network:

“We need one R -band photometric point on this object tonight, good to about a tenth of a magnitude. Can you take it — and how long do you need?”

You have to answer the second question with a number of minutes before you start. The object sets at 01:20 and three other targets are queued behind it, so an open-ended exposure is not an option, and a total that is too short returns a point too noisy to use.

Part 1 is that estimate: **how much telescope time does one photometric point on this star cost?** It is built from seven quantities, taken in order.

What you know when the alert arrives		Where it comes from
Target magnitude	$R = 20.0$	the broker’s latest photometry
Required SNR	10	the coordinator asked for ~ 0.1 mag; by Eq. (10), $\sigma_{\text{mag}} = 1.0857/\text{SNR}$
Filter	R	the network’s light curves are in R
Gain setting	60	the rig’s standard science gain
Sensor temperature	-10°C	what the cooler holds on a September night
Seeing	3.0 arcsec FWHM	an ordinary night at this site
Focal length	300 mm, pixels 5.2 μm	the instrument you are connected to
Sub-exposure cap	300 s	past this the guiding drifts and stars trail

No console. No calculator app you cannot explain. Derive, *in this order*:

- 1.1 Geometry:** the plate scale s , the FWHM in pixels, and the aperture size n_{pix} . (Not one of the seven numbers, but nothing else works without them. n_{pix} is how many pixels’ worth of sky you must measure in order to measure one star.)
- 1.2 The sky-limited floor** $t \geq 10R^2/B$ — the shortest sub worth taking. Below it you pay the read-noise toll on every frame in the stack.
- 1.3 The saturation ceiling** $t \leq 0.7 \text{FW}/(P + B + D)$, taking the target itself as the brightest thing in the aperture.
- 1.4 A sub-exposure length** inside the window those two bounds leave. State the rule you used to choose it, and say *which* of the three limits — floor, saturation, guiding — binds.

- 1.5 SNR **per sub** from Equation (6). Write out all four noise terms separately and give each as a fraction of the total; that tells you what is limiting the observation.
- 1.6 The **number of subs** N and the **total on-sky time**. This is the figure you report back.
- 1.7 The achieved SNR and the resulting σ_{mag} . Compare it with the tenth of a magnitude that was requested.

► **Hand in:** seven numbers — floor, ceiling, sub length, SNR_{sub} , N , total time, σ_{mag} — *with the working*. A number without its working scores nothing. Then complete the reply: “Yes — _____ minutes on sky, _____ subs of _____ s in R , photometry to $\pm$ _____ mag.”

Estimate first. Before working through the arithmetic, write down how many minutes you expect a mag-20 star to need on this telescope for a 0.1 mag point. Compare it with your final answer. Most estimates are out by a factor of several, and the size and direction of the error is worth knowing.

Part 2 — Fly it

45 minutes, on the simulator

- 2.1 Log in to your **workshop** account and complete the pre-flight checklist.
- 2.2 Put Part 1 into the **Exposure** tab: focal length 300 mm, pixel size 5.2 μm , magnitude 20.0, required SNR 10, filter R, gain 60, temperature -10 , seeing 3.0, max sub 300 s. **The planner must agree with your paper.** If it does not, one of you is wrong — find out which, and write down what the disagreement was.
- 2.3 Go to **Candidates**, poll ALerCE, and pick an observable target. Note its class, peak altitude, airmass and moon separation, and say in one line why it is or is not worth a slot tonight.
- 2.4 **Approve** → **Queue**, then **Observing** → **Launch**. Watch the whole chain and name each stage as it happens: slew → plate-solve → centre → autofocus → guide → expose → archive.
- 2.5 Open the archived frame and read its FITS header.

► **Hand in:** from the header — OBJECT, FILTER, EXPTIME, GAIN, EGAIN, READNOIS, CRVAL1, CRVAL2; and from the archive row — HFR and the star count. Plus one screenshot of the execution monitor mid-run.

Where each number lives. OBJECT, FILTER, EXPTIME, GAIN, EGAIN and READNOIS are written into the FITS header at the edge, when the frame is authored. CRVAL1/CRVAL2 are injected *after* a successful plate solve, and only when CRITO_SOLVE_SCIENCE_FRAMES=true. HFR and star count are *not* header keywords at all: they are measured at ingest by ASTAP and stored on the archive row — read them in the console’s Archive list.

Deliverable and marking

One page: **the numbers, the header, one screenshot**. Be ready to state, in two minutes, your total time and *which* of the three limits — sky floor, saturation, guiding — set your sub length.

Part	Marks	What earns them
1	55	Seven correct numbers with working (35); the completed reply (5); naming which bound binds the sub length, by how much, and why (15).
2	45	Planner reconciled with paper, any disagreement diagnosed (15); execution chain named stage by stage (10); header and archive values transcribed correctly (15); candidate justified in one line (5).

4 Hints

Read these only when you are stuck. They are ordered, and each gives away a little more than the last.

Part 1

H1.1. Do the geometry first and write down three numbers before you touch the CCD equation: s in arcsec/px, FWHM in px, and n_{pix} . Every later step uses n_{pix} , and it is the single easiest place to lose a factor of π .

H1.2. $206.265 \times 5.2/300$ is just under 3.6 arcsec/px, so at 3 arcsec seeing the PSF is narrower than one pixel. The instrument is under-sampled — that is a property of a wide-field rig, not an error — and it is why n_{pix} comes out at about five pixels.

Note also that $n_{\text{pix}} \propto \text{seeing}^2$, and that Equation (3) contains no camera property. The seeing, not the sensor, sets how many pixels of background are measured alongside the star. This matters again at step 1.5.

H1.3. For the ceiling you need the PSF *peak* rate, not the total: the star's light is spread over the PSF, and only the central pixel is at risk of saturating. That is Equation (5), and σ there is in *pixels*.

H1.4. The ceiling for a mag-20 star comes out at thousands of seconds. That is not an error. Consider whether saturation is the constraint that actually limits your exposure at this magnitude.

H1.5. A defensible rule for choosing the sub: *three times the sky-limited floor*, clamped into the window. At $3\times$ the floor, read noise is about 3% of your noise budget — comfortably sky-limited, without making each sub so long that one satellite trail costs you a big fraction of the night.

H1.6. N must be a whole number, and you must round *up*. Rounding up means you overshoot the required SNR slightly — report the SNR you actually achieve, not the one you asked for.

Part 2

H2.1. If the planner disagrees with your paper by a large factor, check the focal length and pixel size in the form *first*: they must read 300 mm and 5.2 μm . Selecting a saved setup with different optics replaces both fields without warning, and every result downstream changes with them.

H2.2. If the planner disagrees by a few percent only, you are both right and the difference is rounding — in n_{pix} , or in where you rounded N . Say so explicitly; that is a correct answer, not a failure.

H2.3. Empty candidate list? The broker returned nothing, which is a real operational condition, not a bug. Lower CRITO_ALERCE_PROBABILITY, widen CRITO_ALERCE_LOOKBACK_DAYS, or clear the classifier and re-poll.

H2.4. Nothing launches? Check for a *manual override* badge in the Observing tab. Any manual mount or camera command pauses the queue by design — a human touching the telescope always wins. Clear the override and resume.

H2.5. To read the header without leaving the terminal:

```
curl -s localhost:8000/api/images/<id>/fits -o frame.fits
python -c "from astropy.io import fits; print(repr(fits.open('frame.fits')[0].header))"
```

A Doing it from the command line

The Exposure tab, the CLI and the API all call the same function. Any of these is an acceptable way to do Part 2.

```
# the whole of Part 1, from the calibration table
python -m crito.transient.exposure \
  --calibration calibration/minicam8.example.yaml \
  --mag 20.0 --snr 10 --filter R --gain 60 --temp -10 \
  --focal-length 300 --pixel-size 5.2 --seeing 3 --max-sub 300

# same thing with the constants typed in by hand (no table needed)
python -m crito.transient.exposure \
  --mag 20.0 --snr 10 --read-noise 3.2 --full-well 38000 \
  --sky 3.0 --zp 20.3 --dark 0.01 \
  --focal-length 300 --pixel-size 5.2 --seeing 3 --max-sub 300

# protect a bright field star from saturation (extra credit)
python -m crito.transient.exposure ... --brightest-mag 12

# over HTTP
curl localhost:8000/api/tools/exposure/calibration
curl -X POST localhost:8000/api/tools/exposure \
  -H 'content-type: application/json' \
  -d '{"mag":20.0,"required_snr":10,"filter":"R","gain":60,
      "temp_c":-10,"seeing_arcsec":3,"max_sub_s":300}'

# the transient chain
```

```
curl localhost:8000/api/transient/night      # tonight's dark window
curl -X POST localhost:8000/api/transient/poll # poll ALERCE now
curl "localhost:8000/api/transient/candidates?group_by=class"
```

B Troubleshooting

Symptom	Cause / fix
focal length unknown (400)	no focal_length_mm in observatory.yaml; pass -focal-length or fill the form field
calibration for gain N is incomplete	read_noise_e or full_well_e is null in the table
filter 'X' not in calibration	add it under filters:, or tick <i>manual constants</i>
Exposure tab: <i>no calibration table</i>	CRITO_CALIBRATION_FILE unset or wrong path; restart the backend
Plan wants an absurd total time	sky-limited faint broadband target — that <i>is</i> the answer, not a bug: check n_{pix} and the sky rate
<i>saturation ceiling below sky floor</i>	bright star in a dark sky: lower the gain (more full well) or accept a read-noise penalty
Frames are blank noise	gsc missing from the simulator image; nothing will plate-solve
Nothing solves	FOV hint does not match the star database — the simulated optics are 300 mm/5.2 μm , which wants D05
No CRVAL1/CRVAL2 in the header	CRITO_SOLVE_SCIENCE_FRAMES is false (the default)
Queue will not launch	<i>manual override</i> is latched — a manual command always pauses the queue; clear it and resume
403 on any move command	the account is <code>sim_only</code> and this backend is not a simulator; check the SIMULATION badge
Candidate list empty	the broker returned nothing: lower CRITO_ALERCE_PROBABILITY, widen the lookback, re-poll
Polar alignment always reads ~ 0	the simulated mount is perfectly aligned; the maths is being exercised, not your judgement

C Symbols

Symbol	Meaning	Units
s	plate scale	arcsec/px
θ_{px}	seeing FWHM in pixels	px
σ	Gaussian PSF width, $\theta_{\text{px}}/2.3548$	px
n_{pix}	pixels in the photometry aperture	px
m	target magnitude	mag
ZP	zero point: the magnitude giving $1 \text{ e}^- \text{ s}^{-1}$	mag
S	source count rate	$\text{e}^- \text{ s}^{-1}$
P	PSF peak rate, central pixel	$\text{e}^- \text{ s}^{-1} \text{ px}^{-1}$
B	sky background rate	$\text{e}^- \text{ s}^{-1} \text{ px}^{-1}$
D	dark current	$\text{e}^- \text{ s}^{-1} \text{ px}^{-1}$
R	read noise (<i>not</i> squared)	e^-
FW	full well	e^-
t	sub-exposure length	s
N	number of subs	—
σ_{mag}	1- σ magnitude uncertainty	mag

Built on CRITO. The planner is `crito.transient.exposure`; the constants are `calibration/minicam8.example.yaml`. Replace them with your own measurements before you believe any of it.